

A disproof of Conjecture 00000007752

The Böttcher coordinate of $z^2 - 2$ takes algebraic values at algebraic points

1 The conjecture

Conjecture 00000007752 reads (English version):

Definition: The Bottcher coordinate ϕ_c of $f(z) = z^2 + c$ is the unique univalent analytic function near infinity with $\phi_c(f(z)) = \phi_c(z)^2$. *Conjecture:* For every algebraic c (with 0 non-escaping), the ratio $\phi_c(z)/z$ at algebraic points z of sufficiently large modulus is transcendental; and for two distinct parameters c_1, c_2 , the ratios $\phi_{c_1}(z)/\phi_{c_2}(z)$ are algebraically independent at algebraic points of the common domain.

The Chinese version describes ϕ_c as the unique *single-valued* (dan zhi) analytic function near ∞ with $\phi_c(f(z)) = \phi_c(z)^2$; for every algebraic number c for which 0 does not escape, the ratio of $\phi_c(z)$ to z at algebraic points z with $|z|$ sufficiently large is transcendental; and for $c_1 \neq c_2$ the ratios $\phi_{c_1}(z)/\phi_{c_2}(z)$ are algebraically independent. The formalization follows the English requirement of univalence and the standard meaning of a Böttcher coordinate.

Result. Clause 1 is false for $c = -2$, where $\phi_{-2}(z) = \frac{1}{2}(z + s(z))$ is algebraic. Here $s(z) = z(1 - 4/z^2)^{1/2}$, with the principal root, is the square root of $z^2 - 4$ asymptotic to z at ∞ . It is *not* the principal square root of $z^2 - 4$: the two differ in sign at $z = -4$. Clause 1 is also false for $c = 0$, where $\phi_0(z) = z$. Clause 2 is false for $(c_1, c_2) = (0, -2)$. Every step is a Lean 4/Mathlib theorem (Section 5).

2 Reading

The text leaves three points open. We refute every combination (eight readings of clause 1).

- (R1) “0 *non-escaping*”: either the orbit $f_c^n(0)$ does not tend to ∞ (**ZeroNonEscaping**), or it is bounded, i.e. c is in the Mandelbrot set (**ZeroOrbitBounded**). Our parameters satisfy both.
- (R2) “*Bottcher coordinate*”. The text omits the normalization. Reading (N), **IsBottcherCoord**: ϕ is analytic and injective on $\{|z| > R\}$, $\phi(z^2 + c) = \phi(z)^2$ there, and $\phi(z)/z \rightarrow 1$ as $z \rightarrow \infty$. This is the standard Böttcher coordinate, which conjugates f_c to $w \mapsto w^2$ near ∞ . Reading (U), **IsBottcherSolution**: the text’s own conditions (analytic, injective, functional equation) with “analytic near infinity” read on the Riemann sphere, i.e. $\zeta \mapsto \phi(1/\zeta)$ is meromorphic at 0, and with no normalization. Under (U) the solution is *not* unique: $1/\phi$ is a solution as well (Lean `sol_inv`, `sol2_inv`, `sol0_inv`). This is the “ ϕ versus $1/\phi$ ” ambiguity.
- (R3) *Quantifier*. Since “the unique” fails under (U), we refute both “for every coordinate ϕ ” (**Clause1Forall**) and “for some coordinate ϕ ” (**Clause1Exists**). “Sufficiently large modulus” is read in its weakest form, $\exists R \forall z$ algebraic with $|z| > R$.

The conjecture is the conjunction of clauses 1 and 2. The main theorem refutes clause 1 conjoined with an arbitrary proposition P , so it does not depend on how clause 2 is read. Clause 2 is refuted separately (**Clause2**: the family $z \mapsto \phi_{c_1}(z)/\phi_{c_2}(z)$ over the algebraic z with $|z| > R$ is algebraically independent over $\mathbb{Q}$).

3 Definitions and sources

$f_c(z) = z^2 + c$ (**fc**). Algebraic and transcendental numbers are Mathlib's **IsAlgebraic** $\mathbb{Q}$ and **Transcendental** $\mathbb{Q}$, and algebraic independence is **AlgebraicIndependent** $\mathbb{Q}$. “Near ∞ ” is **nbhdInf** $R = \{z : R < |z|\}$, limits at ∞ use Mathlib's **cobounded** $\mathbb{C}$, and meromorphy is **MeromorphicAt**. Wikipedia (*Böttcher's equation*) states: “Böttcher's coordinate (the logarithm of the Schröder function) conjugates $h(z)$ in a neighbourhood of the fixed point to the function zn .” and “An especially important case is when $h(z)$ is a polynomial of degree n , and $a = \infty$.” The same article lists power maps and Chebyshev polynomials as cases where Böttcher coordinates can be computed explicitly. With $T_2(x) = 2x^2 - 1$ one has $z^2 - 2 = 2T_2(z/2)$, so $z^2 - 2$ is linearly conjugate (via $z = 2x$) to the Chebyshev polynomial T_2 . The normalization $\phi(z)/z \rightarrow 1$ in (N) is the standard one (Milnor, *Dynamics in One Complex Variable*, Thm. 9.1). It is not needed under (U).

4 The proof

Let $\psi(z) = \frac{z}{2}(1 + t(z))$ with $t(z) = (1 - 4/z^2)^{1/2}$ (principal branch; **psi2**). So $\psi(z) = \frac{1}{2}(z + s(z))$, where $s(z) = z t(z)$ is the square root of $z^2 - 4$ asymptotic to z at ∞ (not the principal square root of $z^2 - 4$).

Lemma 1 (**psi2_quad**, **norm_psi2**). *For $z \neq 0$, $\psi(z)^2 - z\psi(z) + 1 = 0$. For all z , $|\psi(z)| \geq |z|/2$.*

Proof. $t^2 = 1 - 4/z^2$ gives $\psi^2 - z\psi + 1 = \frac{z^2}{4}(t^2 - 1) + 1 = 0$. By Mathlib's **cpow_inv_two_re**, $\operatorname{Re} t = \sqrt{(|x| + \operatorname{Re} x)/2} \geq 0$ with $x = 1 - 4/z^2$. So $|1 + t| \geq 1$. $\square$

Lemma 2 (**model2**). *On $|z| > 3$: $\psi(z^2 - 2) = \psi(z)^2$; $\psi(z) \neq 0$; $|z^2 - 2| \geq 2|z|$; and $\psi(z)$ is algebraic whenever z is. Moreover $\psi(z)/z \rightarrow 1$ as $z \rightarrow \infty$.*

Proof. Put $a = z^2 - 2$. Both $\psi(a)$ and $\psi(z)^2$ are roots of $W^2 - aW + 1$. For the second root this follows from $\psi^4 - (z^2 - 2)\psi^2 + 1 = (\psi^2 - z\psi + 1)(\psi^2 + z\psi + 1)$. Both roots have modulus > 1 , because $|\psi(a)| \geq |a|/2 > 1$ and $|\psi(z)|^2 > 9/4$. Two distinct roots of $W^2 - aW + 1$ have product 1, so the two values coincide (**root_unique**). If z is algebraic, so is t^2 , hence t (**IsAlgebraic.of_pow**), hence $\psi(z)$. Finally $\psi(z)/z = (1 + t)/2$, and $t \rightarrow 1^{1/2} = 1$ by continuity of the power function at 1. $\square$

Lemma 3 (**psi2_analytic0n**, **psi2_inj0n**, **psi2_mero**). *ψ is analytic and injective on $\{|z| > 3\}$, and $\zeta \mapsto \psi(1/\zeta) = \zeta^{-1} \cdot \frac{1}{2}(1 + (1 - 4\zeta^2)^{1/2})$ is meromorphic at 0. Hence ψ satisfies both (N) and (U) (**coord2**, **sol2**).*

Proof. For $|z| > 2$ the number $1 - 4/z^2$ has positive real part, so it lies in the slit plane, where $w^{1/2}$ is analytic. Injectivity holds because $z = (\psi^2 + 1)/\psi$. $\square$

Lemma 4 (**Rigidity**; **limit_idem**, **rigid**). *Suppose $|f_c(z)| \geq 2|z|$ and $G(f_c(z)) = G(z)^2$ for $|z| > R$. If $G \rightarrow \ell$ at ∞ , then $\ell^2 = \ell$. If $G \rightarrow 1$, then $G = 1$ on some $\{|z| > R'\}$.*

Proof. Since $f_c \rightarrow \infty$, the limit of $G \circ f_c = G^2$ is both ℓ and ℓ^2 . For the second claim, choose R' so that $|G - 1| < \frac{1}{2}$ on $|z| > R'$, and put $u_k = G(f_c^k z) - 1$. The orbit stays in the region, and $u_{k+1} = u_k(u_k + 2)$ with $|u_k + 2| \geq \frac{3}{2}$. So $\frac{1}{2} > |u_k| \geq (\frac{3}{2})^k |u_0|$ for all k , which forces $u_0 = 0$. $\square$

Theorem 5 (Classification; `Model.classify`, `Model.coord_unique`, `Model.solution_alg`). *Let $c \in \{0, -2\}$ with $\psi = \text{id}$, resp. $\psi = \text{psi2}$. (a) Every ϕ satisfying (N) equals ψ on some $\{|z| > R\}$. So (N) has a unique solution. (b) Every ϕ satisfying (U) is either 0 or ψ^m ($m \in \mathbb{Z}$) on some $\{|z| > R\}$. In both cases $\phi(z)$ is algebraic for every algebraic z with $|z| > R$.*

Proof. (a) Apply the rigidity lemma to $G = \phi\psi^{-1} = (\phi/z)(\psi/z)^{-1} \rightarrow 1$. (b) If the order of $\phi(1/\zeta)$ at 0 is ∞ , then $\phi = 0$ near ∞ . Otherwise $\phi(1/\zeta) = \zeta^n g(\zeta)$ with g analytic and $g(0) \neq 0$. Then $G = \phi\psi^n = (\psi/z)^n g(1/z) \rightarrow g(0)$, and $G \circ f_c = G^2$. The rigidity lemma gives $g(0)^2 = g(0)$, hence $g(0) = 1$, then $G = 1$, i.e. $\phi = \psi^{-n}$. Integer powers of algebraic numbers are algebraic. $\square$

Theorem 6 (`conjecture_7752_false`, `refuted_by_zero`, `clause2_false`). *For both readings (R1), both readings (R2) and both quantifiers (R3), clause 1 fails for $c = -2$ (and for $c = 0$), and so does its conjunction with any P . Clause 2 fails for $(c_1, c_2) = (0, -2)$.*

Proof. $c = -2$ is algebraic, and $0 \mapsto -2 \mapsto 2 \mapsto 2$, so the orbit is bounded and does not escape (`bounded2`, `nonEscaping_of_bounded`). A coordinate exists (`coord2`, `sol2`), and by the classification every coordinate has algebraic values at large algebraic z . For any R , take the integer $z = \lceil R \rceil + 1$, which exceeds R and the classification radius (`exists_big`). Then $\phi(z)/z$ is algebraic, not transcendental (`refute`). For clause 2, $z/\psi(z)$ is algebraic, and every member of an algebraically independent family is transcendental. $\square$

5 Lean 4 formalization

File `lean/Conjecture7752/Basic.lean` (namespace `C7752`, 482 lines, Lean v4.33.1, Mathlib v4.33.1, only import `Mathlib`). Main theorem (ASCII rendering):

```
theorem conjecture_7752_false (P : Prop) :
  forall Hyp in [ZeroNonEscaping, ZeroOrbitBounded],
  forall IsBC in [IsBottcherCoord, IsBottcherSolution],
  not (Clause1Forall Hyp IsBC /\ P) /\ not (Clause1Exists Hyp IsBC /\ P)
```

where `Clause1Forall Hyp IsBC` is $\forall c, \text{IsAlgebraic } \mathbb{Q} \ c \rightarrow \text{Hyp } c \rightarrow \forall \phi, \text{IsBC } c \ \phi \rightarrow \exists R, \forall z, \text{IsAlgebraic } \mathbb{Q} \ z \rightarrow R < \|z\| \rightarrow \text{Transcendental } \mathbb{Q} \ (\phi z/z)$, and `Clause1Exists` has $\exists \phi, \text{IsBC } c \ \phi \wedge \exists R \dots$ instead. Further theorems: `refuted_by_zero` (same conclusion for $c = 0$), `clause2_false`, `Model.coord_unique`, `Model.solution_alg`, `sol2_inv`.

Axiom audit. `lake build` succeeds, and the file also compiles with `-DwarningAsError=true`. `#print axioms for conjecture_7752_false, refuted_by_zero, clause2_false, Model.coord_unique and sol2_inv` reports `[proptext, Classical.choice, Quot.sound]` each. There is no `sorry`, no `native_decide` and no added axiom.

6 Remarks on scope

$c = 0$ and $c = -2$ are the classical explicit cases: the power map, and a map linearly conjugate to the Chebyshev polynomial T_2 (Section 3). A reviewer may regard them as special, but the statement does not exclude them: it says “every algebraic c (with 0 non-escaping)”. Both are in

scope: c is algebraic, 0 is non-escaping, and the coordinate exists and is unique under (N). We do not claim anything about other parameters.

Formal scope. The Lean theorems cover exactly the readings (R1)–(R3) above. In particular, `IsBottcherSolution` *assumes* that ϕ is meromorphic at ∞ (`MeromorphicAt`). The existential variant “some ϕ univalent near ∞ with no condition at ∞ ” is covered only mathematically: univalence on $\{|z| > R\}$ rules out an essential singularity at ∞ , so such a ϕ is meromorphic there and falls under (U). That step is not formalized in Lean. The universal form of this variant needs no such step, because ψ itself satisfies it.